

VIDEO GAME SALES ANALYSIS

Data Cleaning, Exploration and Regional Sales Insights

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5,893	15	2010–2017
Final records	Table Columns	Years covered

1. Background and Objective

This report presents an analysis of a video game sales dataset covering 5,893 game releases sold across the United States and Australia between 2010 and 2017. The dataset tracks national and global sales figures, along with profit, genre, platform, publisher, and regional breakdowns.

Video game sales data offers insight into regional demand, platform popularity, and publisher performance over time. The objective of this analysis is to explore and analyze the video game sales data in order to answer the following questions:

- How are national sales distributed across regions and countries?
- What proportion of total sales (national and global) comes from each country?
- What data quality issues are present in the dataset, and what steps are required to prepare the data for analysis?

2. Data and Methodology

This is a public data obtained from the tutor's github account provided as a CSV file containing 15 fields and 5,909 records. The column names are: Rank, Name, Platform, Year, Month, Genre, Publisher, Country, City, State, Region, National Sales, Global Sales, National Profit, and Global Profit.

The data was read into Pandas in Python using `pd.read_csv()` and all the analysis was done using Jupyter notebook. Other packages used are Numpy, Matplotlib and Seaborn.

2.1 Data Cleaning

The following actions were taken in cleaning the data:

- Duplicate records: 16 duplicate rows were identified and investigated before removing them using `.duplicated()` and `.drop_duplicates()`, reducing the dataset from 5,909 to 5,893 rows.
- Missing values: The Publisher column had 12 missing entries and the Region column had 27 missing entries. Region nulls were filled with the value "North"; Publisher nulls were filled with the placeholder "Unknown".
- Currency formatting: The National Sales column was stored as text with a "\$" prefix (e.g., "\$41.49"). The "\$" symbol was stripped using a regex-based `.replace()` operation.
- Data type correction: After removing the currency symbol, National Sales was still stored as a string. It was converted to a proper numeric type using `pd.to_numeric()` with `errors='coerce'`, changing the column's dtype from str to float64 with zero conversion failures.
- Text standardization: Inconsistent country labels (e.g., "USA") were standardized to "United States" using `.replace()`.
- Column renaming: Columns were renamed for clarity and readability — `NA_Sales` to National Sales, `Global_Sales` to Global Sales, `NA_Profit` to National Profit, and `Global_Profit` to Global Profit.

2.2 Data Manipulation

The data was explored and manipulated to generate insights. Aggregations were performed using `.groupby()` to summarize National Sales by Region and Country, and to summarize both National Sales and Global Sales by Country, in preparation for visualization.

3. Findings

3.1 National Sales by Region and Country

The bar chart below presents total sales by region, comparing United States and Australia.

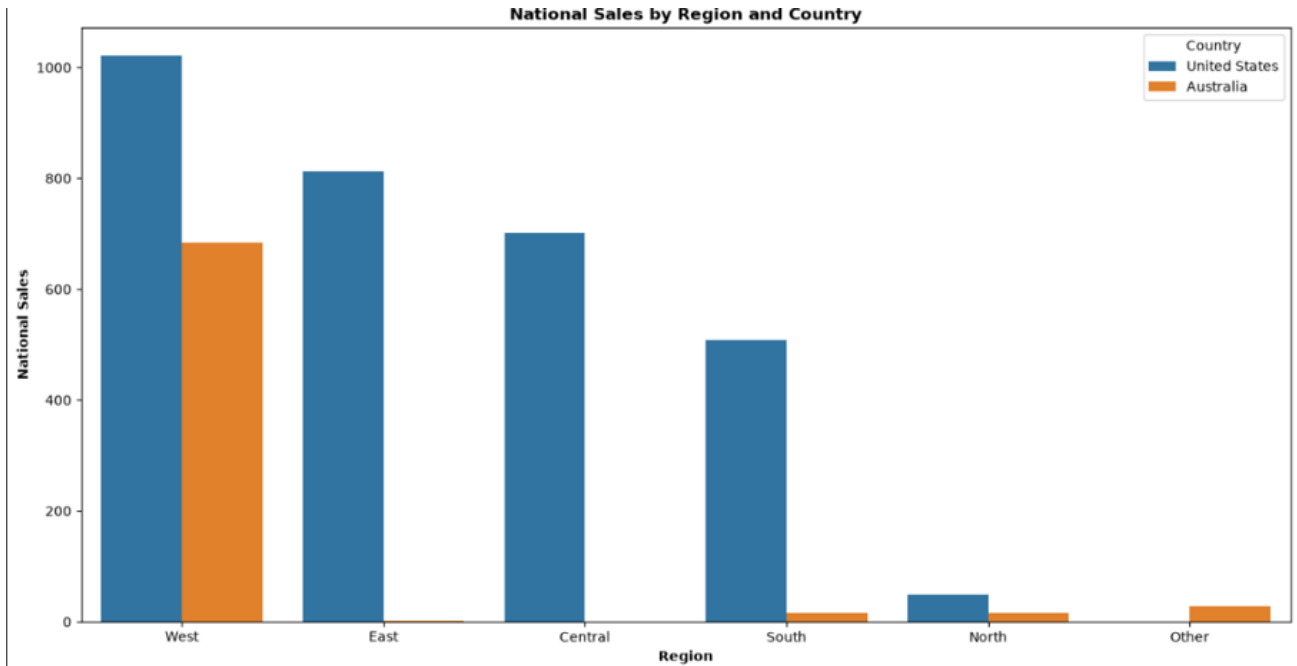


fig 1

The West region generated the highest national sales overall (over 1,000 in national sales units), and was also the only region with substantial contribution from Australia. The East and Central regions were driven almost entirely by United States sales, while the South and North regions showed smaller totals. The "Other" region category consisted exclusively of Australian sales.

3.2 Sales by Country

The pie charts below compare the proportional split of national and global sales between the United States and Australia.

National Sales by Country

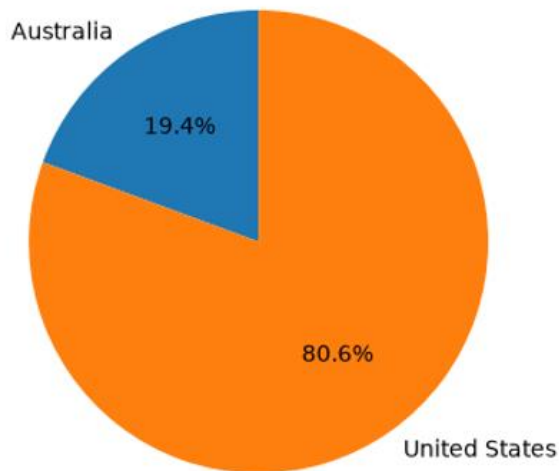


fig 2

Global Sales by Country

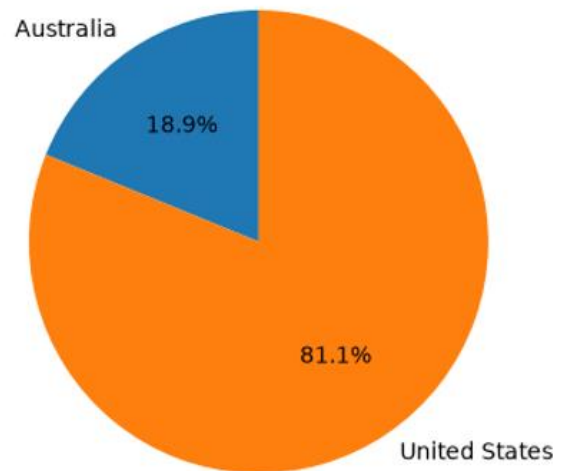


fig 3

The United States accounted for 80.6% of national sales and 81.1% of global sales, with Australia contributing the remaining 19.4% and 18.9% respectively. This near-identical split between national and global figures suggests the two countries in this analysis represent the full scope of sales activity, with the United States as the dominant market by a wide margin.

4. Limitations

- The dataset only contains sales data for two countries (United States and Australia), limiting the ability to draw conclusions about global video game markets more broadly.
- 27 missing Region values were filled with a single default ("North"), which may introduce bias into regional comparisons; the true regional distribution of these records is unknown.
- 12 missing Publisher values were filled with "Unknown" rather than researched and corrected individually, which slightly limits the accuracy of any publisher-level analysis.
- The extreme sales value associated with Wii Sports was retained as a legitimate outlier rather than removed or capped; this may skew simple averages if not accounted for in future analysis (median-based statistics are recommended over means for this dataset).

5. Summary and Recommendations

After the whole analysis (removing 16 duplicate records, correcting data types, standardizing text values, and addressing missing entries), the analysis found that the United States drives the overwhelming majority of both national sales (80.6%) and global sales (81.1%), with Australia accounting for the remaining share. Regionally, the West led in national sales, followed by the East and Central regions.

Sales strategy, marketing spends, and inventory decisions should remain heavily weighted toward the United States market, particularly the West region, while Australia represents a smaller but still meaningful secondary market.

01

Prioritize the United States market, and the West region in particular, in future marketing and inventory planning, given its outsized share of both national and global sales.

02

Investigate the true regional distribution of the 27 records currently defaulted to "North" to improve the accuracy of regional analysis.

03

Research and correct the 12 "Unknown" publisher records where possible, to support more reliable publisher-level performance analysis.

04

Expand the dataset to include additional countries or regions if the goal is to understand global (rather than two-country) sales patterns.

05

Use median-based statistics alongside means when reporting on sales figures, given the confirmed presence of legitimate high-value outliers such as Wii Sports.

7. Appendix

7.1 Dataset Overview

Attribute	Detail
Original row count	5,909
Final row count (after de-duplication)	5,893
Columns	15
Countries covered	United States, Australia
Years covered	2010 - 2017
Missing values (Publisher)	12 (filled with "Unknown")
Missing values (Region)	27 (filled with "North")

7.2 Tools Used

- Python 3.13
- Pandas (data cleaning and manipulation)
- NumPy (numeric operations)
- Matplotlib and Seaborn (visualization)
- Jupyter Notebook (VS Code)